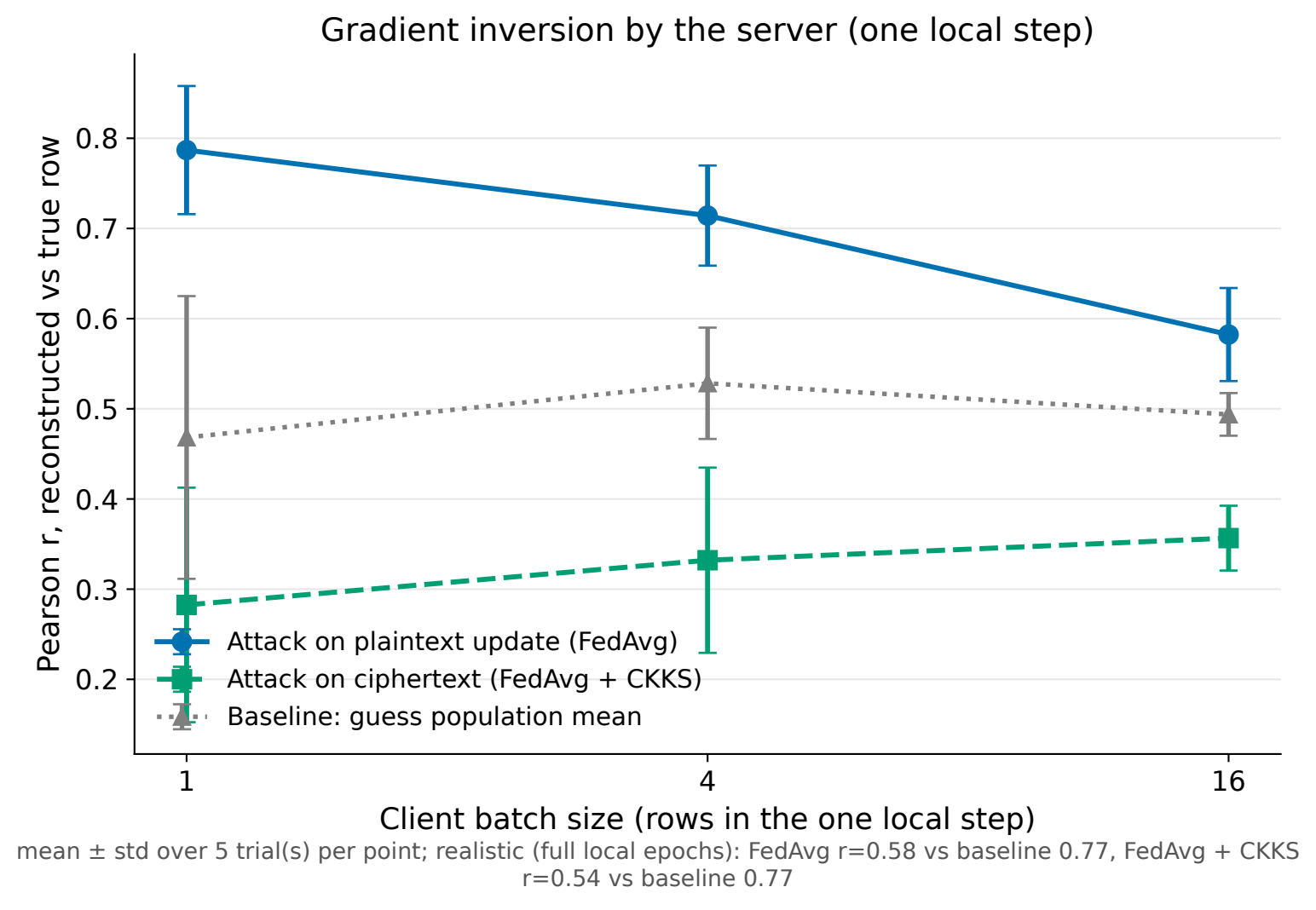
Gradient inversion by the server (one local step)
Pearson r, reconstructed vs true row
0.8
0.7
0.6
0.5
0.4
0.3
0.2
Attack on plaintext update (FedAvg)
Attack on ciphertext (FedAvg + CKKS)
Baseline: guess population mean
1
4
16
Client batch size (rows in the one local step)
mean ± std over 5 trial(s) per point; realistic (full local epochs): FedAvg r=0.58 vs baseline 0.77, FedAvg + CKKS r=0.54 vs baseline 0.77